



(C)lock In

Cassidy Dailly

Introduction

The functions of (C)lock In are designed to improve wake-up routine efficiency while also keeping users focused during study times.

(C)lock In functions as a normal alarm clock, designed to give users a more natural and healthy wake-up routine. The alarm clock is set as normal, but a lock function allows you to store your phone away before sleeping. Science has shown that using your phone first thing in the morning can be detrimental to your productivity, anxiety, and phone-dependency. This being known, the alarm clock is designed to withhold the user's phone for 30 minutes after the alarm is disabled.

Additionally, the clock implements LED brightness to mimic sunrise before wakeup. Abrupt alarms with no lead up are disruptive to our natural clock, making our sleep more restless. Instead, a prolonged change to the environment prior to the alarm noise mimics our body's natural wake up with the sun. To portray this, the physical clock includes back-lit acrylic that will increase in brightness beginning 20 minutes prior to the set alarm time. With the combined light and phone-locking functions, the clock would aid in more productive and healthy wake-up routines.

For the focus function, (C)lock In is designed to keep distractions away from the users by placing physical barriers as blocks to phone access. The times for keeping the phone away would be set using the pomodoro method, which the user would be able to personalize prior to locking their phone away. When the time for each pomodoro round is up (25 minutes), the box unlocks and a timer begins for a break (5 minutes).

(C)lock In is designed as a phone lock box, which would require a certain amount of time to "unlock" the phone. This would be used for both the focus and alarm clock functions, using science to identify time periods in which phones should be kept away. Additionally, if the user wants to unlock their phone early, they are required to complete a challenge or game. This ensures that the phone would be accessible in emergencies while still keeping a barrier present.

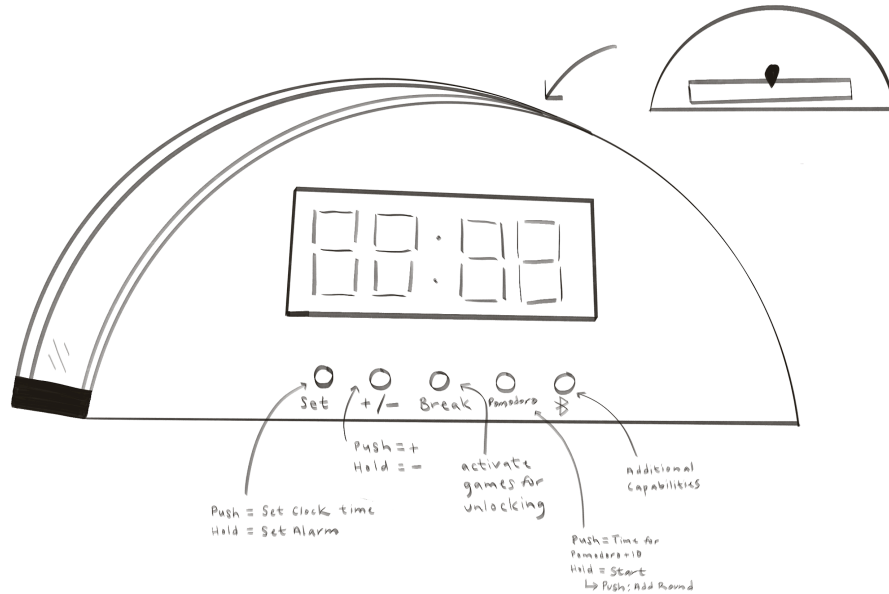


Figure 1. Original (C)lock In Design

Components and Construction

The microcontrollers used in this project were an Arduino Mega and an ESP32-S3. The Arduino was used for screen control because of its adaptability to the original, shield-type screen. The ESP was chosen due to its dual core capabilities, which allowed for a smoothly run alarm clock with multiple interlocking functions.

For the locking mechanism, it was decided to use a push-pull solenoid with a relay that will lock when power is not connected and will open when power is supplied. The [solenoid](#) being used requires 12V, so it is connected to an external source made up of 8 AA batteries. The connection between the solenoid and batteries will be made with a relay, which will take a GPIO output to open and close the connection and therefore open and close the lock. The typical high output of the ESP32-S3 had to be taken into account, and the relay had to be wired with 3.3V instead of 5V to match the GPIO pins.

The LED strip used was from Adafruit and used the Adafruit_NeoPixel.h API to alter the brightness. The testing implementation maps the brightness from 0 to 255. The LED starts incrementing 25 minutes prior to the alarm, increasing by 10 every minute. The option to do

smaller incrementations (which would be preferred) is also there, though this was saved for future implementation because of computation and power usage. The alarm used for the clock was a simple buzzer, and allowed for an extremely awakening noise to be amplified through the use of a transistor. While the natural noise of the speaker was too low, using a transistor allowed the buzzer to amplify in its surrounding.

The screen used was an OLED screen rented from the Van Leer parts studio. The original plan was to use a TFT LED 3.5" screen, however two broken screens resulted in this smaller black and white option. The writing and font of the screen was controlled with a library called U8g2lib. The screen was connected to an Arduino Mega through SPI, and the Mega used UART communication protocols to communicate with the ESP32. The mega also did all of the time and countdown computation. Because of this, the two microcontrollers send data packets back and forth in a call-response method, where different functions are encoded with different characters.

The IR remote had to be connected and mapped based on the inputs to the different buttons. This was used to avoid the need for many push buttons. The IR remote is used for all user interaction, including setting time and alarms, unlocking the lock box, and playing the game. With 21 buttons, a lot of opportunities are opened for device functionality and future expansions. This remote used the IRremote.hpp library.

The clock function is integrated with the Mega. The <TimeLib.h> API runs like an RTC, so the seconds are incremented continuously. When the Mega senses that a new minute has occurred, it updates its screen and sends a data packet to the ESP32 with the new time.

RFID was used as a user input, allowing for the "Home Base" function of the clock. The RFID took a lot of time to implement, but allows the user to place a "base" away from their bed/workstation so that turning the alarm off requires movement. The base was made using an RFID card, which was implanted to line up with an RFID sensor implanted into the bottom of the clock. These devices used the MFRC522.h library and SPI to read radio frequencies.

The microcontroller used RTOS to run two cores at the same time, so the IR was continuously being read even when other clock actions were happening. This is particularly useful for when the Arduino messages the ESP. The cores are split into clock_basics and clock_computations. The screen is also running with the clock and the timer.

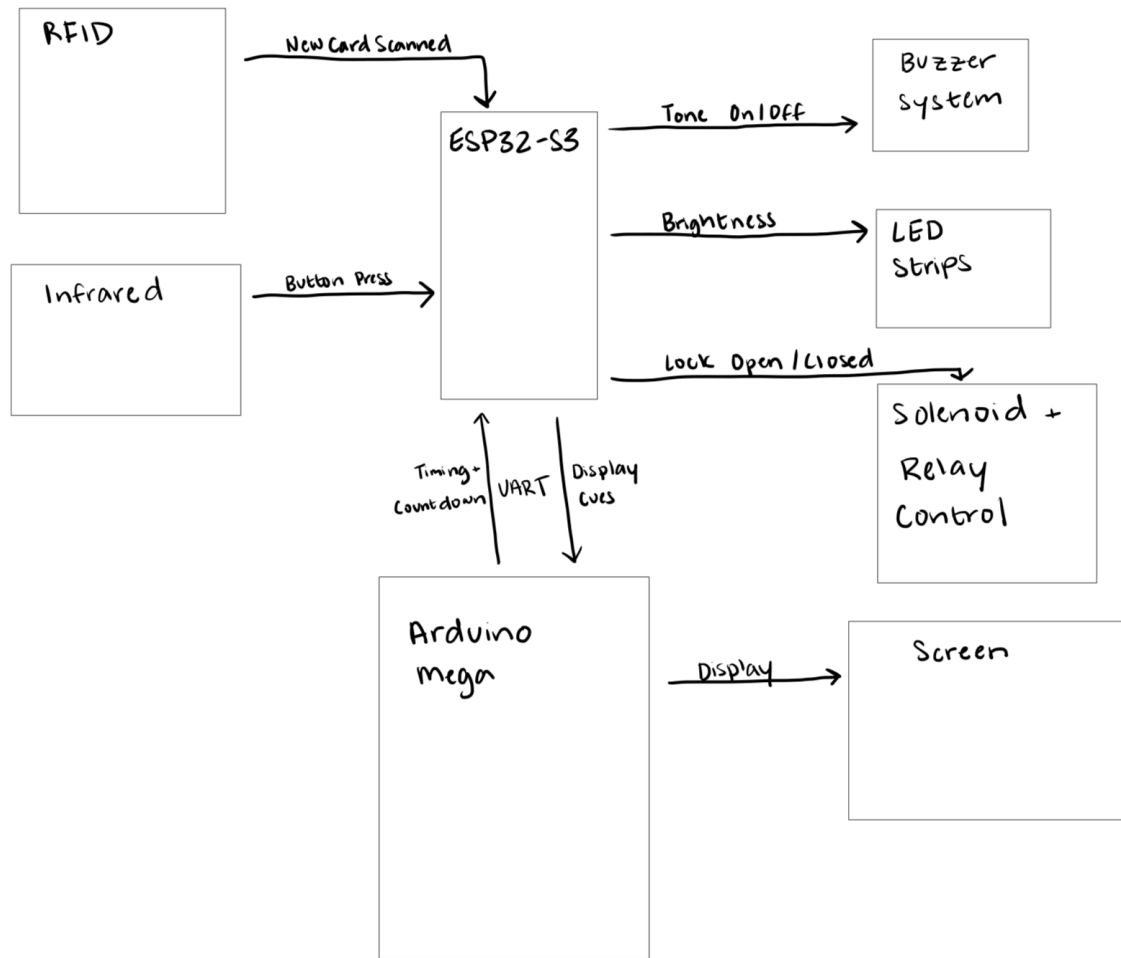


Figure 2. Component Diagram

Process

The project was designed as a retro style alarm clock with backlit acrylic and a lockbox in the back. While it went through multiple design iterations, the premise stayed the same throughout the whole process. Because time constraints and resources altered the project through different stages, there still remain many different alterations or additions to the project that will be discussed later.

The material for the physical clock was chosen to make the construction of the box simple while still allowing for aesthetics. Acrylic was chosen because of tutorials seen and the interest that followed in trying to recreate said projects. It also gave a unique way for light to be shown from the clock itself. The back box was originally planned to be a door, but because of the small size and inaccuracies of 3D

printing, the hinges were very difficult to get right. This resulted in a drawer for the lockbox eventually being printed and added to the back along with a laser cut back wall made from thin wood. Additionally, sides of the back panel had to be cut due to the inability to power the Arduino with the available batteries. Because the screen required so much power, the arduino needed to be plugged into an external power source such as a power bank. While it would have been preferred to use a power bank with dual adapters to power both the ESP and Arduino, the lack of access to one that met these qualifications resulted in the project remaining tethered to outside devices. This could be avoided with a different portable charger than the one obtained, which was one output.

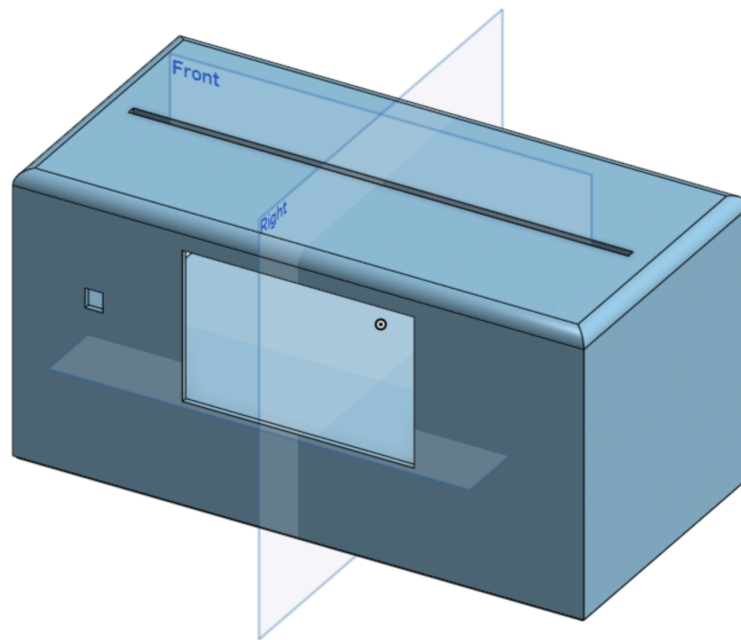


Figure 3. (C)lock In box design

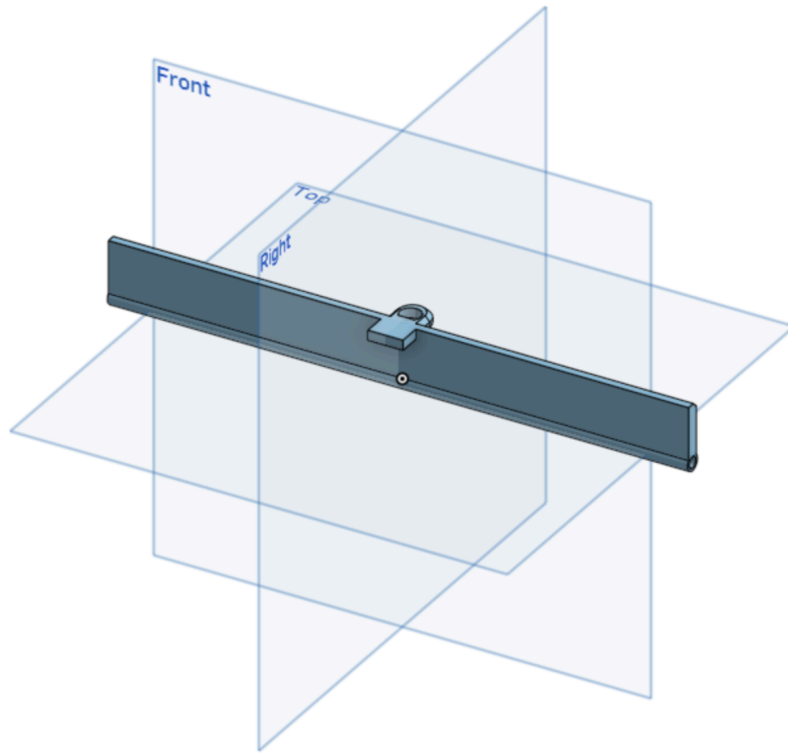


Figure 4. Original lock door design, before changing to drawer

The development of the project progressed in phases. After the original planning and designing, the components were individually tested and coded to be integrated in the end. Changes made from the original design include the use of a transistor with a speaker, the use of a relay to control the solenoid, and the use of RFID as a way to unlock the lock box.

Originally, the plan was to use a transistor to control the solenoid power flow, but this didn't work out due to the high voltage being unfit for the transistors at hand. Other options considered were a MUX and a power control module, but the relay was chosen because of availability and ease of use. The mechanical switching between power and no power was able to provide confidence that a burn-out would not occur.

The infrared led to design choices being made, specifically because of its odd shape and need to be visible for the remote. In the end, the IR receiver was attached to a PCB and placed on the front of the clock. This let the remote change the clock from many visible positions, which was helpful. One challenge faced with the infrared was that the receiver started to go stale towards the end of testing the project, hinting that a real design would need a better quality sensor with less need for replacement.

Additionally, another plan was to make one of the alarm disarming actions a function of the IMU. However, RFID was chosen for two reasons: 1) it didn't require immense shake protection and 2) it

helped the user meet a goal in the morning, which benefits the goal of improving morning routines. The RFID was implemented into a golden base, which adds to the aesthetic of the clock while also giving the product complexity for at-home use. The greatest challenge came in coding the RFID, which was difficult to reset when applied to the project's entirety. Once triggered, the RFID had to be completely erased before it could be triggered again. This took a lot of trial and error before the code was able to continuously run this function.

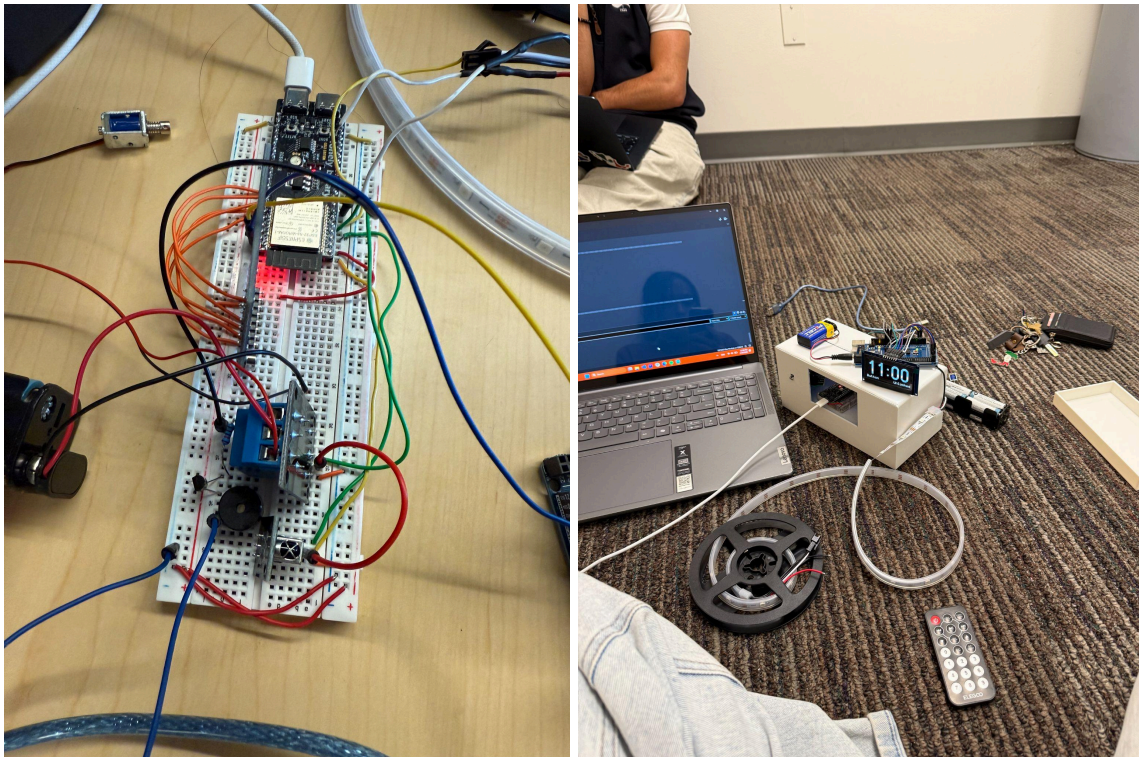


Figure 5. Project build up before integration

An extra design decision that was made was to only open the solenoid box in 10 second increments. Because of the size of the solenoid and the voltage, the metal would conduct quite fast and result in a concerning amount of heat for continuous function. To aid in avoiding solenoid burn-outs or massive losses of power, the unlock function was implemented to only open for ten seconds at a time, allowing the user to open and close the box as they need to without pulling unnecessary power.

The design was finished by sanding the pieces 2-3 times each and using 2-3 layers of colored spray paint and primer. They were then coated with a clear coat, which allowed for a more professional, production-like look. The goal was to make the product aesthetic for homes, and this goal was met with a sophisticated color scheme and a strategic mix of wood and painted finish.

Conclusion

This project was an immense learning process, requiring the integration of mechanical design, electrical systems, and software engineering into one. Throughout the development of (C)lock In, I gained experience with a wide range of techniques, including soldering, 3D printing structural components, and laser cutting acrylic and wood for both functional and aesthetic purposes. These skills are highly valuable for future use in projects, allowing ideas to be rapidly prototyped, tested, and refined into working systems.

One of the most important lessons learned during this project was the necessity of planning for hardware failure. After encountering two broken LCD screens close to the deadline, it became clear that ordering multiple components is essential to maintaining progress and avoiding risks.

This project also emphasized the need for design planning prior to body development. Because of the size of the box, the placement of different electrical components was particularly hard to manage. While all the components eventually did fit, there was a great struggle in getting everything to work cohesively in such a cramped environment. Especially with last-minute changes in power sources, the product didn't allow for a lot of adjustment, so it would have been helpful to have more planned out from the start.

Beyond technical skills, this project also strengthened abilities to iterate on designs and adapt to constraints. Many initial ideas, such as the larger TFT display or hinged door mechanism, had to be modified due to practical limitations. Learning to pivot effectively while maintaining the core functionality of the project was a key takeaway that will be applicable to future engineering work.

Finally, this project showed the improved capabilities from the semester, and the new abilities gained through coursework and hands-on labs. Many of the skills learned were put to the use, and the real-world application allowed for this new knowledge to be tested and improved upon through trial and error. Not many examples were used, and if they were used it was simply to find libraries for components, and this was the most time consuming and rewarding part of the process. While the project took many hours of hard work, it was a great learning experience on learning itself.

Looking forward, there are many opportunities to expand and improve (C)lock In. Future implementations could include Bluetooth or WiFi connectivity to allow automatic time synchronization and remote configuration. The audio system could be upgraded to support higher-quality speakers capable of playing user-selected songs instead of buzzer tones. Additional interactive features, such as games displayed on an upgraded screen, could add to user engagement while still supporting the product's behavioral goals. Finally, incorporating adjustable LED color settings would allow users to customize

their wake-up environment further, potentially improving comfort and effectiveness or just enhancing the aesthetics of the design.

Overall, this project not only achieved its goal of creating a functional and behavior-focused alarm system, but also provided valuable experience in product hardware design, problem-solving, and iterative engineering.